Unet

https://github.com/milesial/Pytorch-UNet

https://arxiv.org/abs/1505.04597
https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28

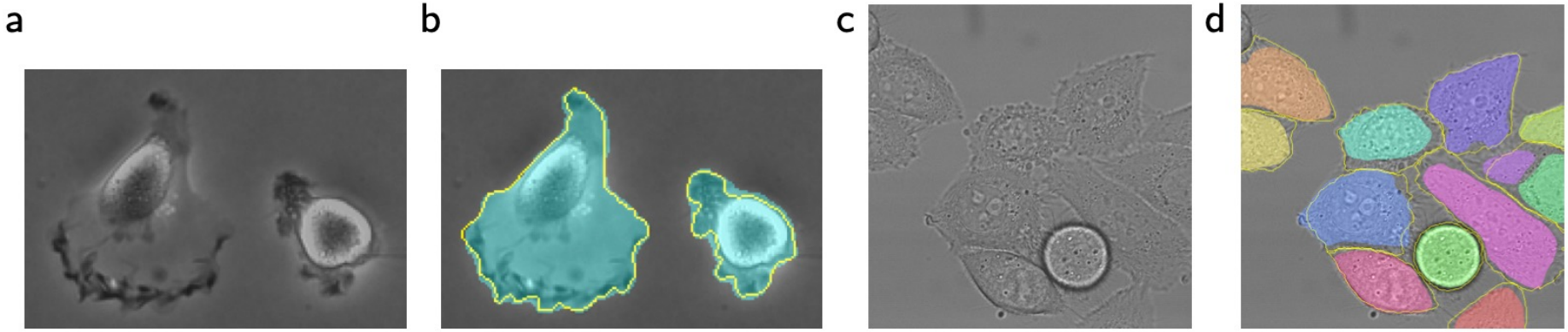
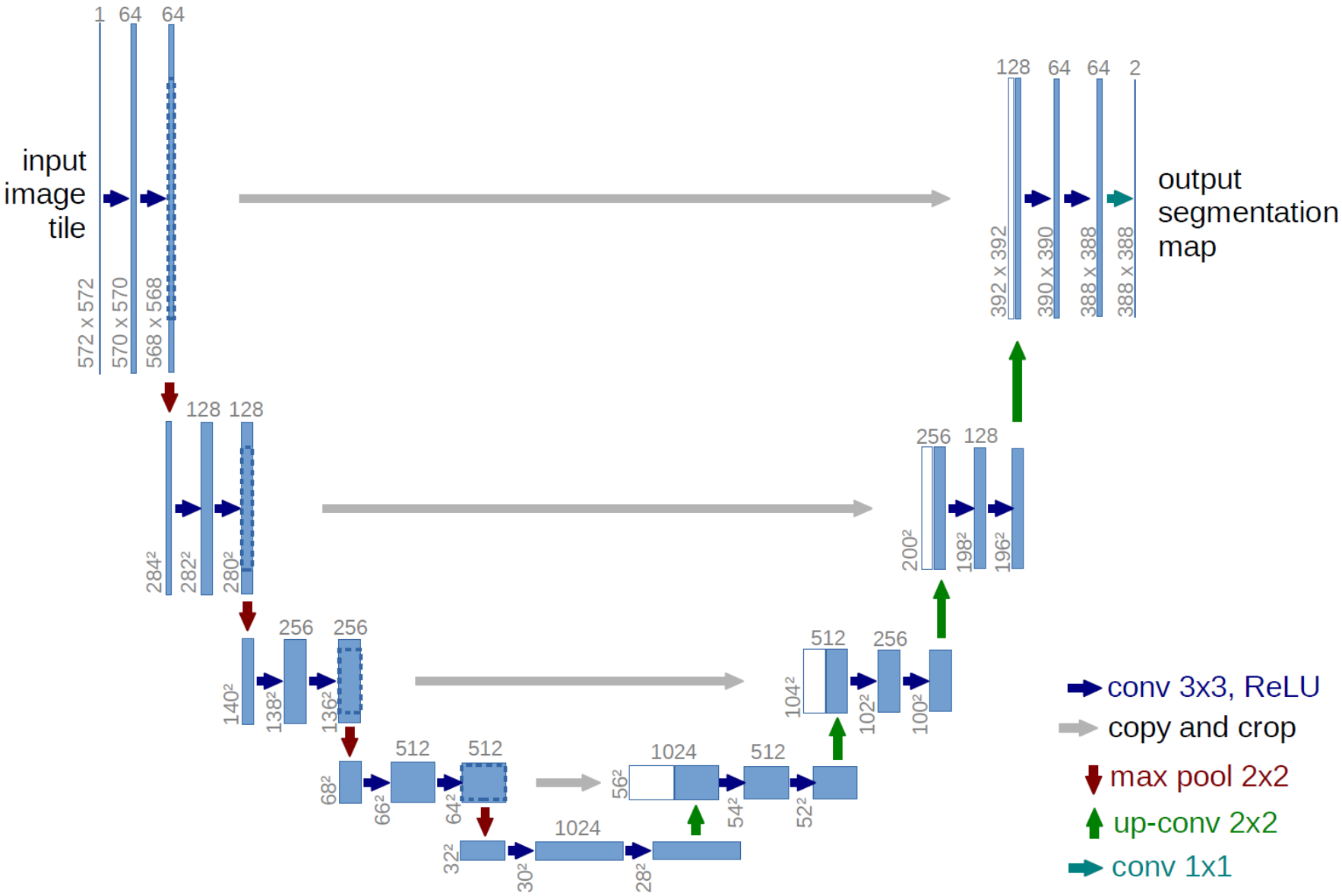
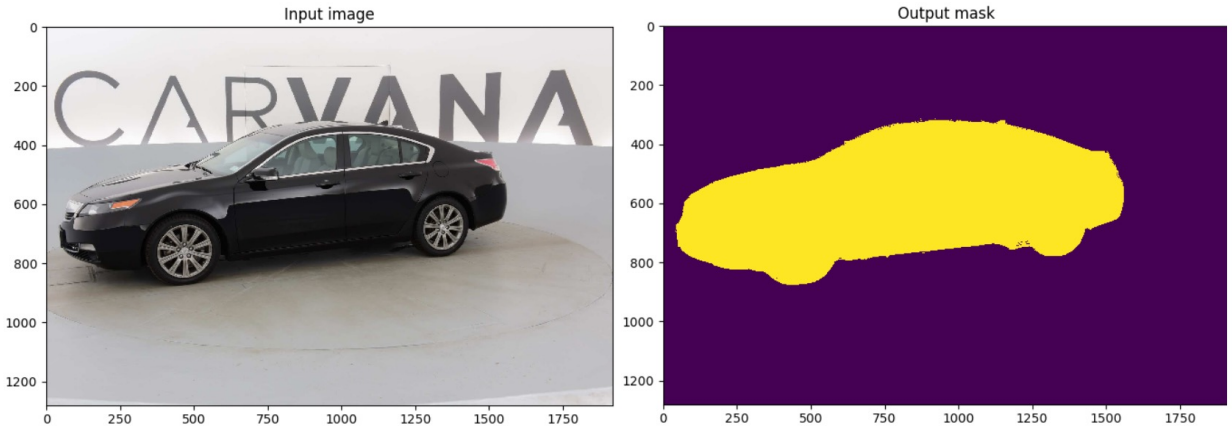


Fig. 4. Result on the ISBI cell tracking challenge. (**a**) part of an input image of the "PhC-U373" data set. (**b**) Segmentation result (cyan mask) with manual ground truth (yellow border) (**c**) input image of the "DIC-HeLa" data set. (**d**) Segmentation result (random colored masks) with manual ground truth (yellow border).


input image tile
output segmentation map
1 64 64
572 x 572
570 x 570
568 x 568
128 128
284²
282²
280²
256 256
140²
138²
136²
512 512
68²
66²
64²
1024
32²
30²
28²
1024 512
56²
54²
52²
512 256
104²
102²
100²
256 128
200²
198²
196²
128 64 64 2
392 x 392
390 x 390
388 x 388
388 x 388
conv 3x3, ReLU
copy and crop
max pool 2x2
up-conv 2x2
conv 1x1

U-Net: Semantic segmentation with PyTorch





Customized implementation of the U-Net in PyTorch for Kaggle's Carvana Image Masking Challenge from high definition images.

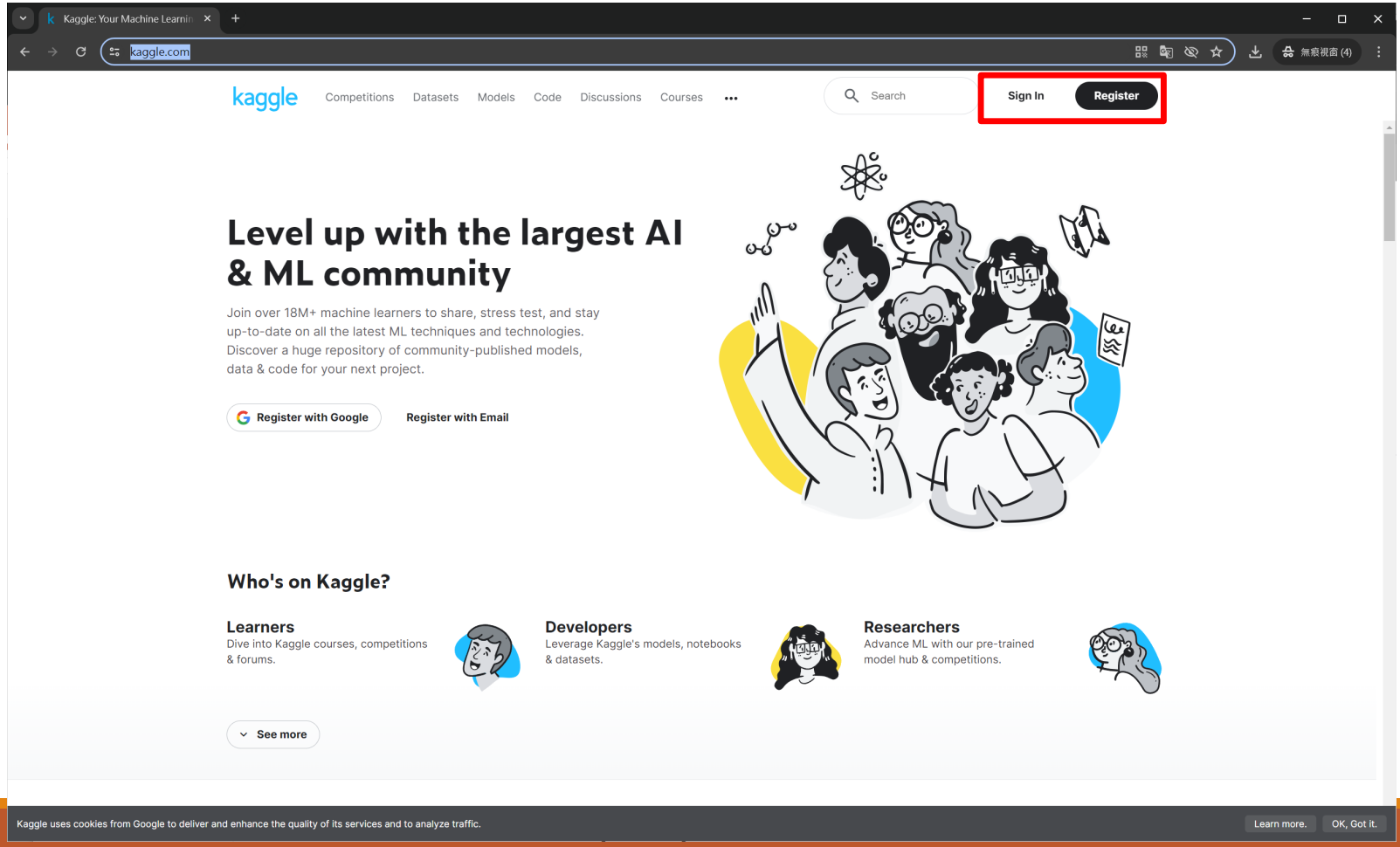
- Quick start
  - Without Docker
  - With Docker
- Description
- Usage
  - Docker
  - Training
  - Prediction
- Weights & Biases
- Pretrained model
- Data

https://github.com/milesial/Pytorch-UNet

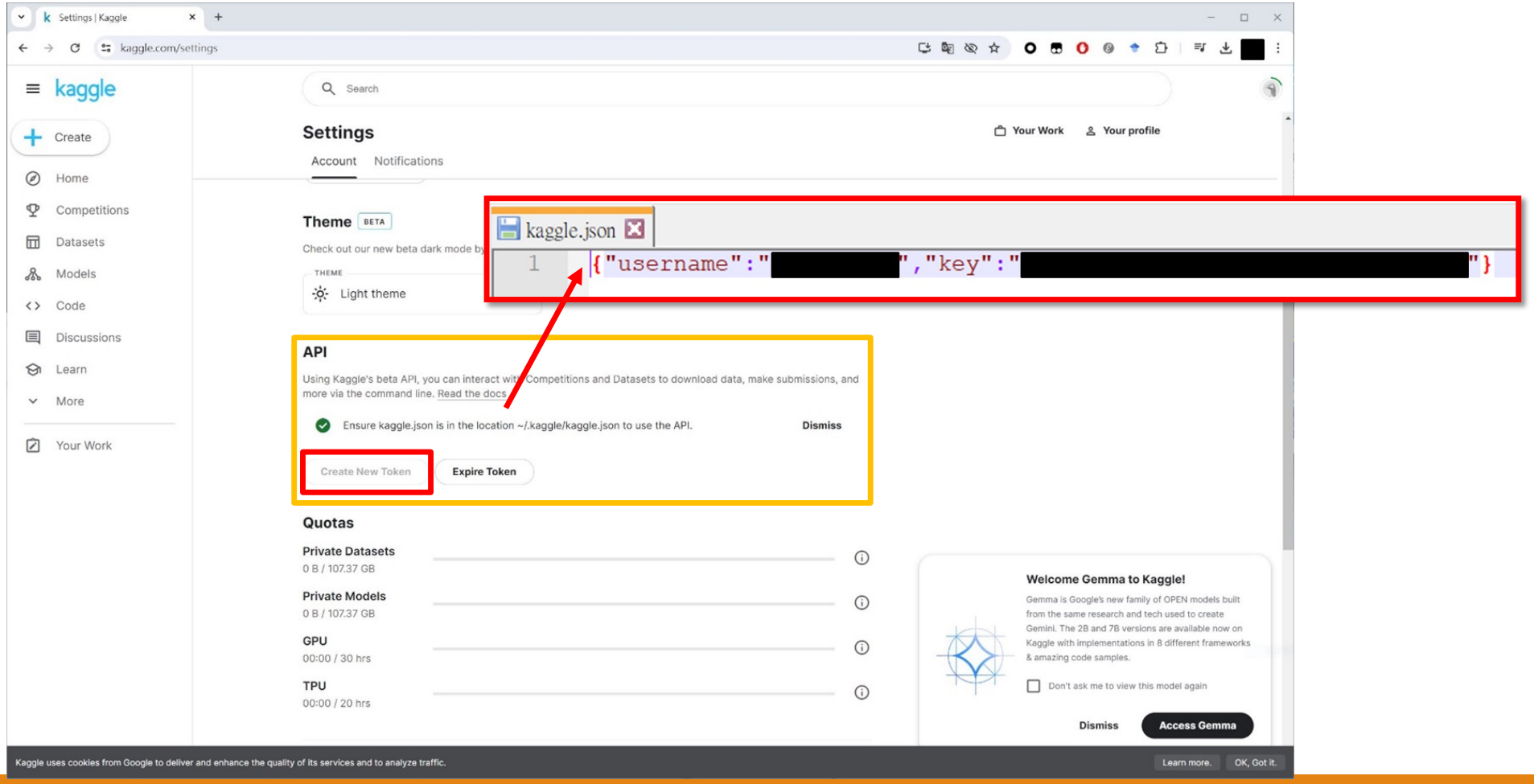
Kaggle API

Kaggle register

https://www.kaggle.com/

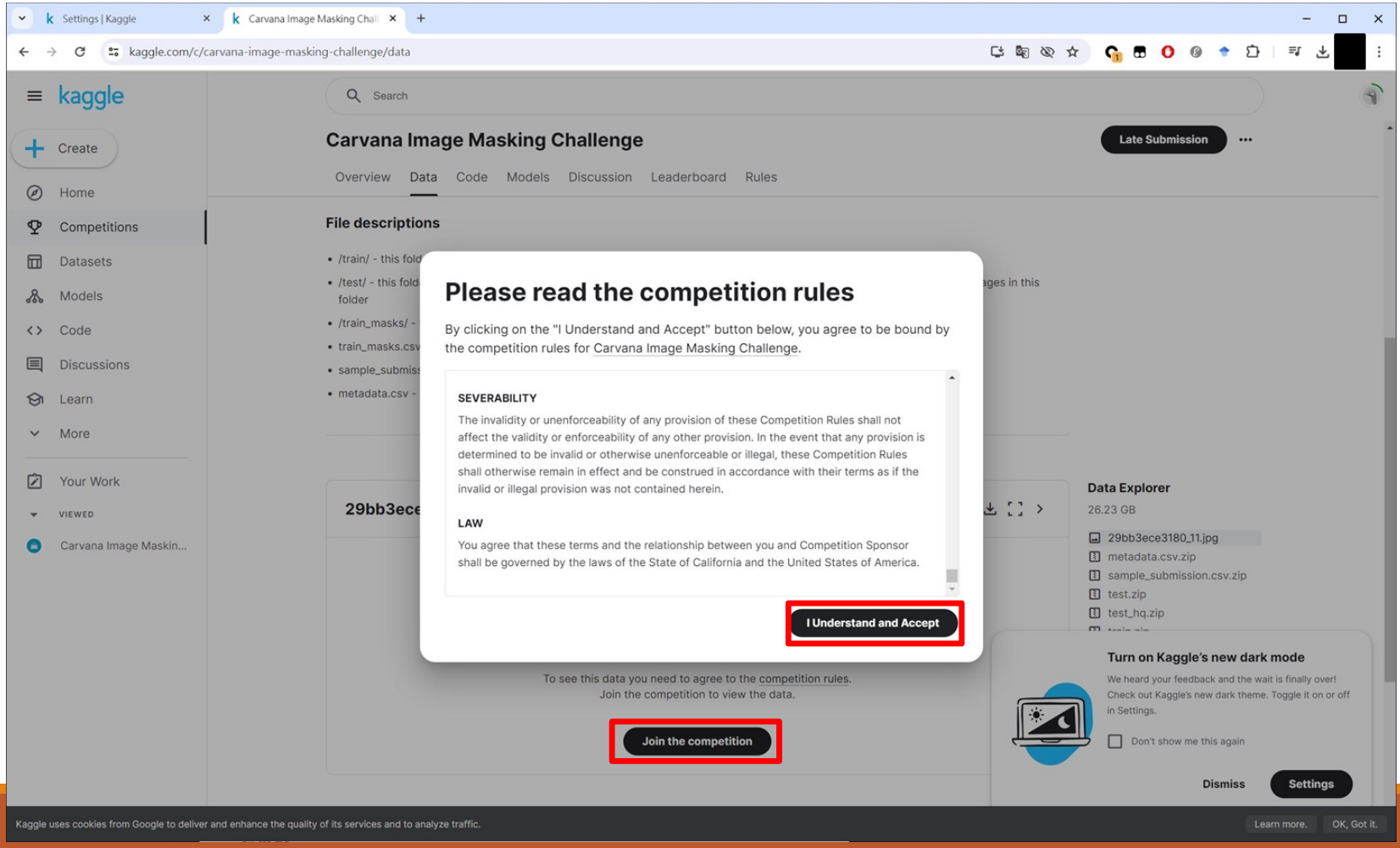


Get Kaggle API

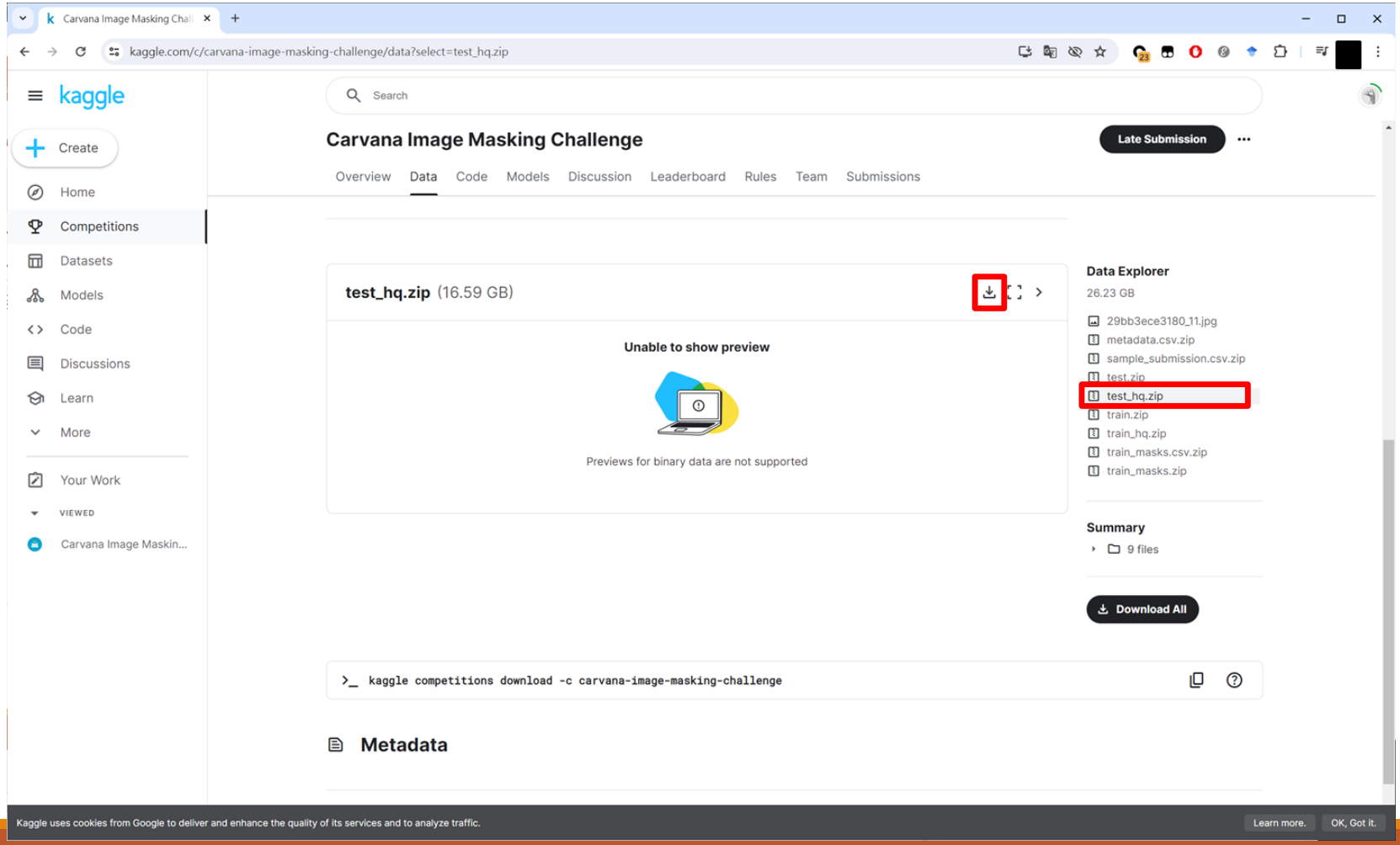


Kaggle參加比賽獲取資料集取得權限

https://www.kaggle.com/c/carvana-image-masking-challenge/data



Download (by URL)



Download (by API)

Input username and key

方法2: 下載kaggle測試資料集-test_hq.zip(16.59 GB)

9分鐘

```
!bash  scripts/download_data2.sh
```

串流輸出內容已截斷至最後 5000 行。

download_data2.sh

```
#!/bin/bash

if [[ ! -f ~/.kaggle/kaggle.json ]]; then
    echo -n "Kaggle username: "
    read USERNAME
    echo
    echo -n "Kaggle API key: "
    read APIKEY

    mkdir -p ~/.kaggle
    echo "{\"username\":\"$USERNAME\",\"key\":\"$APIKEY\"}" > ~/.kaggle/kaggle.json
    chmod 600 ~/.kaggle/kaggle.json
fi

pip install kaggle --upgrade

kaggle competitions download -c carvana-image-masking-challenge -f test_hq.zip
unzip test_hq.zip

```

Exercise:

Observing the classification of the training dataset, collect 10 images with classifications from the training dataset. Each image should contain more than one object or category. Then, execute Unet segmentation. (從訓練資料集觀察分類，請收集包含多個物件或類別的 10 張圖片及其分類。執行 Unet 圖像分割)

Submission requirements:

1. PDF documents
    1. Show the images you collected.
    2. Show the outputs.
2. Upload to e-learning before 5/17 14:10